From Tangled Ropes to the Hecke Algebra: Entropy, Knot Invariants, R-Matrices, and Local Rules

Robin Langer

June 2026

Abstract

These notes trace a thread from the physical question of why ropes tangle to the Iwahori–Hecke algebra, quantum groups, and the local rules appearing in the bijective proof of Borodin's identity for cylindric plane partitions. The central observation is that Fomin's local rules—used in the author's PhD thesis to construct weight-preserving bijections on cylindric plane partitions—are the combinatorial shadow of R-matrices from the quantum inverse scattering method. The commutation relation between Pieri operators and their adjoints plays the role of the Yang–Baxter equation, and the higher-order function formulation of local rules is the combinatorial analogue of the consistency condition that R-matrices must satisfy.

Contents

1 Why ropes tangle **1**
- 1.1 Entropy and conformational space . . . 1
- 1.2 Length dependence . . . 1
- 1.3 Knot classification . . . 2
- 1.4 Why coiling reduces tangling . . . 2

2 Knot invariants and the Jones polynomial **3**
- 2.1 Can you count knots? . . . 3
- 2.2 Braids and Markov's theorem . . . 3

3 The Iwahori–Hecke algebra **3**
- 3.1 Definition . . . 3
- 3.2 Origin: representations of finite groups of Lie type . . . 4
- 3.3 Structure . . . 4

4 Jones's construction **4**
- 4.1 Step 1: Braids into the Hecke algebra . . . 4
- 4.2 Step 2: The Markov trace . . . 4
- 4.3 Step 3: The invariant . . . 4
- 4.4 The surprise: subfactors . . . 5

5 Quantum groups and the Yang–Baxter equation **5**
5.1 Integrable systems and the Yang–Baxter equation 5
5.2 Quantum groups 5
5.3 Quantum Schur–Weyl duality 6
5.4 From Jones to HOMFLY 6

6 Hall–Littlewood polynomials and the Hecke algebra **6**
6.1 Kazhdan–Lusztig theory 6
6.2 Kostka–Foulkes polynomials 6
6.3 Spherical functions on p-adic groups 7

7 Local rules as R-matrices **7**
7.1 Fomin's local rules 7
7.2 Vertex operators and commutation relations 7
7.3 The commutation relation as an R-matrix 8
7.4 The (q, t)-deformation 8
7.5 Local rules as higher-order functions 8
7.6 The correspondence 9
7.7 Connection to knot invariants 9
7.8 Combinatorial R-matrices in crystal theory 9

8 The canonical commutation relations **10**

9 Summary **10**

1 Why ropes tangle

1.1 Entropy and conformational space

Consider a rope of length L confined in a box of side length D. The rope explores *conformational space*—the set of all spatial configurations consistent with its constraints (fixed length, self-avoidance, bending stiffness). The key combinatorial observation is:

> There are astronomically more knotted configurations than unknotted ones.

Unknottedness is a *global* topological constraint: no strand passes through any other along the entire length. Knottedness is *local*: a single crossing change can create a knot. By entropy—the system evolves toward the macrostate with the most microstates—a rope exploring configurations at random will almost certainly land in a knotted state.

1.2 Length dependence

Raymer and Smith [1] tumbled ropes of various lengths in boxes and found:

- Below $\sim$0.46 m: almost no knots form.
- Above $\sim$1.5 m: knot probability approaches 100%.

- The transition is sigmoidal with a critical length regime.

The probability of remaining unknotted decays exponentially:

$$P(\text{unknotted}) \sim e^{-L/L_0}$$

where L_0 is a characteristic length depending on rope stiffness and confinement. This is a standard result in the statistical mechanics of polymers.

The mathematical reason: a rope of length L in a box of size D has roughly L/D segments that can cross over each other. The number of possible crossing diagrams grows exponentially with the number of crossings n, since each crossing can be an over-crossing or under-crossing independently, giving $\sim 2^n$ distinct topological states.

1.3 Knot classification

Raymer and Smith did not merely detect the presence of knots—they *classified* them. After photographing each string configuration, they traced the crossing diagram and computed knot invariants (specifically the Jones polynomial and HOMFLY polynomial) to identify the knot type: trefoils (3_1), figure-eights (4_1), cinquefoils (5_1), and composites up to 11 crossings.

The trefoil was the most common. More complex knots appeared with longer strings. The distribution of knot types roughly matched predictions from random topology models.

1.4 Why coiling reduces tangling

A coiled rope tangles far less than a loosely piled rope of the same length. This is not merely because coiling "holds the rope in place" but because:

1. **Reduced effective length.** A coiled rope behaves topologically like a short, thick rope. The relevant length for knot formation is the number of coils, not the length of rope.

2. **Increased bending stiffness.** Each coil constrains local curvature, pre-spending the curvature budget on orderly loops.

3. **Restricted conformation space.** Coiling imposes a low-entropy initial condition. The energy barrier to transitioning from "neatly coiled" to "randomly tangled" is much higher than from "loosely piled" to "randomly tangled."

4. **Suppressed crossing number.** For N loops, the maximum crossing number scales as $\sim N$ rather than $\sim N^2$.

The same mathematics governs why circular DNA tangles far less than linear DNA of the same length—cells exploit this via topoisomerases.

2 Knot invariants and the Jones polynomial

2.1 Can you count knots?

Given a tangled rope, one can ask: *how knotted is it?*

Just as integers factor uniquely into primes, knots factor uniquely into *prime knots* (Schubert, 1949). A rope with "two knots in it" is a *composite knot*—the connected sum of two prime knots—and the decomposition is unique. So one can count prime knot factors, and the count is well-defined.

However, detecting whether a given tangle is knotted at all is computationally hard. Unknot recognition is known to be in both NP and co-NP, but it is not known to be in P. In practice one computes invariants:

Invariant	**Distinguishes**	**Limitation**
Crossing number	Minimum crossings needed	Hard to compute
Jones polynomial	Most small knots	Unknown if it detects unknot
HOMFLY polynomial	Strictly stronger than Jones	Can't distinguish all knots
Khovanov homology	Detects the unknot	Exponentially expensive

No single invariant distinguishes all knots.

2.2 Braids and Markov's theorem

Every knot can be represented as a *closed braid* (Alexander, 1923). A braid on n strands is an element of the *braid group* B_n, generated by $\sigma_1, \ldots, \sigma_{n-1}$ with relations:

$$\sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1} \qquad \text{(braid/Yang–Baxter relation)} \tag{1}$$

$$\sigma_i \sigma_j = \sigma_j \sigma_i \qquad \text{for } |i-j| \geq 2 \tag{2}$$

Two closed braids represent the same knot if and only if they are related by a sequence of *Markov moves*: conjugation within B_n and stabilisation between B_n and B_{n+1}.

To build a knot invariant, one needs a function on braids invariant under both Markov moves.

3 The Iwahori–Hecke algebra

3.1 Definition

The symmetric group S_n is generated by adjacent transpositions $s_i = (i, i{+}1)$ satisfying $s_i^2 = 1$, the braid relation, and the commutativity relation. The *Iwahori–Hecke algebra* $H_n(q)$, over $\mathbb{C}(q)$, is generated by $T_1, \ldots, T_{n-1}$ satisfying the same braid and commutativity relations, but with $s_i^2 = 1$ replaced by the *quadratic relation*:

$$T_i^2 = (q-1)T_i + q \tag{3}$$

or equivalently $(T_i - q)(T_i + 1) = 0$. At $q = 1$ this reduces to $T_i^2 = 1$, recovering $\mathbb{C}[S_n]$.

3.2 Origin: representations of finite groups of Lie type

The algebra $H_n(q)$ was not invented for knot theory. It arises naturally in the representation theory of finite groups of Lie type (Iwahori, 1964).

Take $G = \mathrm{GL}_n(\mathbb{F}_q)$ and B the Borel subgroup (upper triangular matrices). Consider the permutation representation of G on the flag variety G/B. Then:

$$H_n(q) \cong \mathrm{End}_G\big(\mathrm{Ind}_B^G \mathbf{1}\big) \tag{4}$$

The Hecke algebra is the endomorphism algebra of the induced trivial representation. The parameter q is the cardinality of the finite field.

3.3 Structure

$H_n(q)$ has dimension $n!$ and, for generic q, is semisimple with irreducible representations indexed by partitions $\lambda \vdash n$—the same indexing as for S_n. The q-analogue of the Specht module S^λ gives a representation of $H_n(q)$ of the same dimension.

This is why $H_n(q)$ is called a q-deformation of the symmetric group: it has the same representation theory, parametrised by q.

4 Jones's construction

4.1 Step 1: Braids into the Hecke algebra

There is a natural homomorphism $B_n \to H_n(q)$ sending $\sigma_i \mapsto T_i$. This works because the braid relations are a subset of the Hecke relations. The quadratic relation makes T_i invertible:

$$T_i^{-1} = q^{-1} T_i - (1 - q^{-1})$$

4.2 Step 2: The Markov trace

Jones (1984) found a unique family of linear functionals $\mathrm{tr} : H_n(q) \to \mathbb{C}(q, z)$ satisfying:

1. $\mathrm{tr}(1) = 1$,
2. $\mathrm{tr}(ab) = \mathrm{tr}(ba)$ (trace property),
3. $\mathrm{tr}(h\,T_n) = z \cdot \mathrm{tr}(h)$ for all $h \in H_n(q)$ (Markov property).

Property (3) ensures invariance under the second Markov move (stabilisation).

4.3 Step 3: The invariant

For a braid $\beta \in B_n$ whose closure is the knot K:

$$V_K(t) = (\text{normalisation factor}) \cdot \mathrm{tr}(\rho(\beta))$$

where $\rho : B_n \to H_n(q)$ sends $\sigma_i \mapsto T_i$. After a change of variables, this gives the *Jones polynomial* $V_K(t) \in \mathbb{Z}[t^{\pm 1/2}]$.

Example 4.1. *Some values:*

Unknot:	$V(t) = 1$
Trefoil 3_1 :	$V(t) = -t^{-4} + t^{-3} + t^{-1}$
Figure-eight 4_1 :	$V(t) = t^2 - t + 1 - t^{-1} + t^{-2}$

4.4 The surprise: subfactors

Jones was not studying knots. He was studying *subfactors*—inclusions $N \subset M$ of von Neumann algebras. The Jones index $[M : N]$ turned out to be quantised: it can only take values in $\{4\cos^2(\pi/n) : n \geq 3\} \cup [4, \infty)$.

While analysing the tower of relative commutants, Jones noticed the generators satisfied exactly the Hecke algebra relations. The trace came from the conditional expectation in the von Neumann algebra. A topologist colleague (Joan Birman) pointed out this could yield a knot invariant.

The discovery path was:

Operator algebras → Subfactors → Hecke algebra → Braid representations → Knot invariant

5 Quantum groups and the Yang–Baxter equation

5.1 Integrable systems and the Yang–Baxter equation

The Hecke algebra and Jones polynomial are part of a larger story rooted in physics. The *Leningrad school* (Faddeev, Sklyanin, Takhtajan, Kulish, Reshetikhin) developed the *quantum inverse scattering method* for solving quantum integrable systems (spin chains, lattice models).

The master equation of quantum integrability is the *Yang–Baxter equation*: for $R \in \mathrm{End}(V \otimes V)$,

$$R_{12}\, R_{13}\, R_{23} = R_{23}\, R_{13}\, R_{12} \tag{5}$$

in $\mathrm{End}(V \otimes V \otimes V)$.

This equation was discovered independently by Yang (1967, solving the 1D delta-function Bose gas) and Baxter (1972, solving the 8-vertex model in statistical mechanics).

5.2 Quantum groups

The central question was: *where do R-matrices come from?*

Drinfeld (1985) and Jimbo (1985) independently answered this by defining *quantum groups*—Hopf algebra deformations $U_q(\mathfrak{g})$ of universal enveloping algebras of semisimple Lie algebras. The simplest example is $U_q(\mathfrak{sl}_2)$, with generators E, F, K, K^{-1} and relations:

$$KEK^{-1} = q^2 E \tag{6}$$

$$KFK^{-1} = q^{-2} F \tag{7}$$

$$EF - FE = \frac{K - K^{-1}}{q - q^{-1}} \tag{8}$$

where $K = q^H$. At $q = 1$ one recovers the classical algebra.

The crucial structure is the *coproduct* $\Delta : U_q \to U_q \otimes U_q$:

$$\Delta(K) = K \otimes K \tag{9}$$

$$\Delta(E) = E \otimes 1 + K \otimes E \tag{10}$$

$$\Delta(F) = F \otimes K^{-1} + 1 \otimes F \tag{11}$$

This coproduct is *not cocommutative*—the failure of cocommutativity is controlled by the R-matrix, which satisfies Yang–Baxter automatically.

5.3 Quantum Schur–Weyl duality

Classical Schur–Weyl duality says:

$$\mathrm{End}_{\mathrm{GL}(V)}(V^{\otimes n}) \cong \mathbb{C}[S_n]$$

Deforming $\mathrm{GL}(V)$ to $U_q(\mathfrak{gl}_N)$ deforms $\mathbb{C}[S_n]$ to $H_n(q)$:

$$\mathrm{End}_{U_q(\mathfrak{gl}_N)}(V^{\otimes n}) \cong H_n(q)$$

The Hecke algebra is not an arbitrary q-deformation—it is the *unique* deformation forced by quantum group symmetry.

5.4 From Jones to HOMFLY

The Jones polynomial uses $U_q(\mathfrak{sl}_2)$. Replacing $\mathfrak{sl}_2$ with $\mathfrak{sl}_N$ gives the *HOMFLY polynomial* $P(a, z)$, satisfying the skein relation:

$$a\,P(L_+) - a^{-1}P(L_-) = z\,P(L_0)$$

The Jones polynomial is $P(t^{-1}, t^{1/2} - t^{-1/2})$; the Alexander polynomial is $P(1, t^{1/2} - t^{-1/2})$.

6 Hall–Littlewood polynomials and the Hecke algebra

6.1 Kazhdan–Lusztig theory

Kazhdan and Lusztig (1979) constructed a distinguished basis $\{C'_w\}$ for $H_n(q)$ indexed by permutations $w \in S_n$. The change-of-basis coefficients are the *Kazhdan–Lusztig polynomials* $P_{x,w}(q)$.

6.2 Kostka–Foulkes polynomials

The *Kostka–Foulkes polynomials* $K_{\lambda\mu}(t)$ are the transition matrices between Hall–Littlewood polynomials and Schur functions:

$$s_\lambda(x) = \sum_\mu K_{\lambda\mu}(t)\, P_\mu(x; t)$$

Lusztig (1981) showed that these are special cases of Kazhdan–Lusztig polynomials for Grassmannian permutations. Thus the combinatorics of Hall–Littlewood polynomials is secretly Hecke algebra combinatorics.

6.3 Spherical functions on p-adic groups

Hall–Littlewood polynomials also arise as *spherical functions* on p-adic groups via the Satake isomorphism. For $G = \mathrm{GL}_n(\mathbb{Q}_p)$ with maximal compact $K = \mathrm{GL}_n(\mathbb{Z}_p)$, the Hecke algebra of K-bi-invariant functions on G is isomorphic to the ring of symmetric polynomials, and the spherical functions are precisely $P_\lambda(x; q^{-1})$.

7 Local rules as R-matrices

We now connect the preceding material to the local rules appearing in the author's PhD thesis [8], which gave a bijective proof of Borodin's identity for cylindric plane partitions.

7.1 Fomin's local rules

Fomin's local rules [7] provide a mechanism for filling in a growth diagram. Given three partitions ρ, μ, ν and a bit $x \in \{0, 1\}$ arranged as:

$$\begin{array}{ccc} \rho & & \nu \\ & x & \\ \mu & & \lambda \end{array}$$

the local rule determines the fourth partition λ.

Abstractly, a local rule is a map with type signature

$$\mathfrak{U}_{\alpha,\beta} : (\mathbb{Z}_{\geq 0}, D(\alpha) \cup D(\beta)) \to U(\alpha) \cup U(\beta) \tag{12}$$

where $U(\mu)$ and $D(\mu)$ denote the sets of partitions obtainable from μ by adding or removing a horizontal strip, respectively.

7.2 Vertex operators and commutation relations

In the author's thesis, cylindric plane partitions with profile $\pi = (\pi_0, \pi_1, \ldots, \pi_T)$ are encoded algebraically via *vertex operators*:

$$G^0(z) = \Omega[Xz] \qquad \text{(Pieri operator: adds horizontal strip)} \tag{13}$$

$$G^1(z) = \Omega^*[Xz] \qquad \text{(dual Pieri operator: removes horizontal strip)} \tag{14}$$

The generating function for cylindric plane partitions is then expressed as the trace of a product:

$$\sum_{c \in \mathrm{CPP}(\pi)} z^{|c|} = \sum_{\mu} \langle S_\mu \mid G^{\pi_0}(u_0)\, G^{\pi_1}(u_1) \cdots G^{\pi_T}(u_T)\, D_w\, S_\mu \rangle$$

The key algebraic fact is the **commutation relation**:

$$\boxed{\Omega^*[Xu]\,\Omega[Xv] = \frac{1}{1-uv}\,\Omega[Xv]\,\Omega^*[Xu]} \tag{15}$$

7.3 The commutation relation as an R-matrix

An R-matrix $R \in \mathrm{End}(V \otimes V)$ governs what happens when two objects are swapped—two tensor factors, two braid strands, two operators. Equation (15) has precisely this structure:

- The operators G^0 and G^1 are the "objects" being swapped.
- Commuting them past each other produces a scalar factor $\frac{1}{1-uv}$.
- This scalar factor *is* the R-matrix (in its simplest form: a scalar R-matrix for a 1-dimensional representation).

7.4 The (q,t)-deformation

In the Macdonald polynomial setting, the commutation relation becomes (see [10, 6]):

$$\Omega^*_{q,t}[Xu]\,\Omega_{q,t}[Xv] = \frac{(tuv;q)_\infty}{(uv;q)_\infty}\,\Omega_{q,t}[Xv]\,\Omega^*_{q,t}[Xu] \tag{16}$$

where $(a;q)_\infty = \prod_{n\geq 0}(1-aq^n)$.

This is explicitly a vertex operator commutation relation from the quantum inverse scattering method—the same Jimbo who co-invented quantum groups is a co-author of the relevant reference [6]. The factor $(tuv;q)_\infty/(uv;q)_\infty$ is a q-deformed R-matrix scalar.

Specialisations:

- $q = t$: reduces to $\frac{1}{1-uv}$, recovering the Schur case.
- $q = 0$: gives the Hall–Littlewood case.
- General (q,t): the full Macdonald case.

7.5 Local rules as higher-order functions

In the author's thesis, the local rule was formulated as a *higher-order function*:

$$\mathfrak{L}_i : \big((\mathrm{P}, \mathrm{ALCD}(\pi)) \to \mathrm{CPP}(\pi)\big) \to \big((\mathrm{P}, \mathrm{ALCD}(\pi')) \to \mathrm{CPP}(\pi')\big) \tag{17}$$

where π' differs from π by swapping the entries at positions i and $i+1$. That is, $\mathfrak{L}_i$ takes a weight-preserving bijection ψ_π and returns a new weight-preserving bijection $\psi_{\pi'}$.

Each application of $\mathfrak{L}_i$ corresponds algebraically to commuting $G^{\pi_i}(u_i)$ past $G^{\pi_{i+1}}(u_{i+1})$, picking up the R-matrix scalar factor. The full bijective proof of Borodin's identity is obtained by composing local rules to sort the profile—each step is an adjacent transposition, and the product of scalar factors yields the generating function.

7.6 The correspondence

Combinatorial (local rules)	**Algebraic (R-matrices)**
Vertex operators G^0, G^1	Components of a vertex operator algebra
Commutation relation (15)	R-matrix exchange relation
Local rule $\mathfrak{L}_i$	Combinatorial R-matrix
Composing local rules to sort π	Factoring a permutation into s_i's
Scalar factor $\frac{1}{1-u_i u_j}$	R-matrix eigenvalue
(q,t)-deformation (16)	Quantum group R-matrix
Cylindric plane partition profile π	Tensor product labelling
Growth diagram	Path in the representation graph

7.7 Connection to knot invariants

The parallel with the Jones polynomial construction is now visible:

Jones polynomial	**Local rules / Borodin**
Braid group B_n	Symmetric group S_n (profile permutations)
$\sigma_i \mapsto T_i$ (Hecke generator)	$s_i \mapsto \mathfrak{L}_i$ (local rule)
Braid relation $\sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1}$	Yang–Baxter consistency of local rules
Markov trace $\to$ knot invariant	Composition of local rules $\to$ bijection
Quadratic relation $(T_i - q)(T_i + 1) = 0$	Commutation relation (15)

In both cases, a complex global structure (a knot invariant, or a bijection between combinatorial objects) is built from simple local swaps, each controlled by a commutation/exchange relation. The Hecke algebra is the common algebraic framework.

7.8 Combinatorial R-matrices in crystal theory

The local rules are closely related to the *combinatorial R-matrix* studied by Shimozono, Schilling, and others in the context of Kirillov–Reshetikhin crystals:

- A cylindric plane partition is a highest-weight element in a tensor product of crystal bases.
- The profile π determines which crystals are tensored.
- The local rule $\mathfrak{L}_i$ is the combinatorial R-matrix that swaps two adjacent crystal factors.
- The Yang–Baxter equation guarantees that the result is independent of the order of swaps—which is exactly what ensures the bijection is well-defined.

Borodin's identity is a partition function for a solvable lattice model, and solvable lattice models are precisely the systems governed by R-matrices satisfying Yang–Baxter.

8 The canonical commutation relations

There is a further layer of structure. Let p_1^* denote the adjoint of multiplication by $p_1 = h_1$ with respect to the Hall inner product. The pair (p_1, p_1^*) satisfies the *canonical commutation relations*:

$$[p_1^*, p_1] = 1$$

This can be understood from the fact that an integer partition always has one more outside corner box than inside corner box.

Fomin's original local rule is a bijective proof of the identity:

$$\langle S_\alpha \mid p_1^* p_1 \, S_\beta \rangle = 1 + \langle S_\alpha \mid p_1 \, p_1^* \, S_\beta \rangle$$

which is the matrix element version of $[p_1^*, p_1] = 1$.

The Robinson correspondence itself can then be understood algebraically via the identity $n! = \sum_\lambda (f^\lambda)^2$, which follows from the commutation relations by computing:

$$\begin{aligned} \sum_\lambda f_\lambda f_\lambda &= \sum_\lambda \langle 1 \, p_1^n \mid S_\lambda \rangle \langle S_\lambda \mid p_1^n \, 1 \rangle \\ &= \langle 1 \mid (p_1^*)^n (p_1)^n \mid 1 \rangle \\ &= n! \end{aligned}$$

where the last step uses $[p_1^*, p_1] = 1$ repeatedly.

This places the Robinson correspondence—and with it the entire edifice of growth diagrams and local rules—squarely within the framework of operator algebras and commutation relations.

9 Summary

Tangled ropes $\xrightarrow{\text{classify}}$ Jones polynomial $\xrightarrow{\text{Markov trace}}$ Hecke algebra $H_n(q)$ $\xrightarrow{q\text{-deformation}}$ Quantum groups $\xrightarrow{\text{R-m}}$

The Hecke algebra $H_n(q)$ sits at the intersection of:

- **Topology:** braids, knots, the Jones and HOMFLY polynomials.
- **Algebra:** representations of $\mathrm{GL}_n(\mathbb{F}_q)$ and p-adic groups, Kazhdan–Lusztig theory.
- **Combinatorics:** symmetric functions, Hall–Littlewood and Macdonald polynomials, Kostka–Foulkes polynomials.
- **Physics:** quantum inverse scattering, integrable lattice models, vertex operator algebras.

The fact that tangled ropes, Hall–Littlewood polynomials, and cylindric plane partitions are all controlled by the same algebraic structure is not a coincidence. It is because all three are governed by what happens when you q-deform the symmetric group.

References

[1] D. M. Raymer and D. E. Smith, *Spontaneous knotting of an agitated string*, Proc. Natl. Acad. Sci. USA **104** (2007), 16432–16437.

[2] V. F. R. Jones, *A polynomial invariant for knots via von Neumann algebras*, Bull. Amer. Math. Soc. **12** (1985), 103–111.

[3] D. Kazhdan and G. Lusztig, *Representations of Coxeter groups and Hecke algebras*, Invent. Math. **53** (1979), 165–184.

[4] V. G. Drinfeld, *Quantum groups*, Proc. ICM Berkeley (1986), 798–820.

[5] M. Jimbo, *A q-difference analogue of $U(\mathfrak{g})$ and the Yang–Baxter equation*, Lett. Math. Phys. **10** (1985), 63–69.

[6] M. Jimbo and T. Miwa, *Solitons and infinite-dimensional Lie algebras*, Publ. RIMS, Kyoto Univ. **19** (1983), 943–1001.

[7] S. Fomin, *Generalized Robinson–Schensted–Knuth correspondence*, Zap. Nauchn. Sem. LOMI **155** (1988), 156–175.

[8] R. Langer, *Cylindric plane partitions, Lambda determinants, Commutants in semicircular systems*, PhD thesis, Université Paris-Est, 2013.

[9] A. Borodin, *Periodic Schur process and cylindric partitions*, Duke Math. J. **140** (2007), 391–468.

[10] A. M. Garsia, M. Haiman, and G. Tesler, *Explicit plethystic formulas for Macdonald q,t-Kostka coefficients*, Sém. Lothar. Combin. **42** (1999), Art. B42m.

[11] I. G. Macdonald, *Symmetric Functions and Hall Polynomials*, 2nd ed., Oxford University Press, 1995.

[12] E. Witten, *Quantum field theory and the Jones polynomial*, Comm. Math. Phys. **121** (1989), 351–399.

[13] N. Yu. Reshetikhin and V. G. Turaev, *Ribbon graphs and their invariants derived from quantum groups*, Comm. Math. Phys. **127** (1990), 1–26.